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AI in Social Media

Systems, objectives, evidence and governance

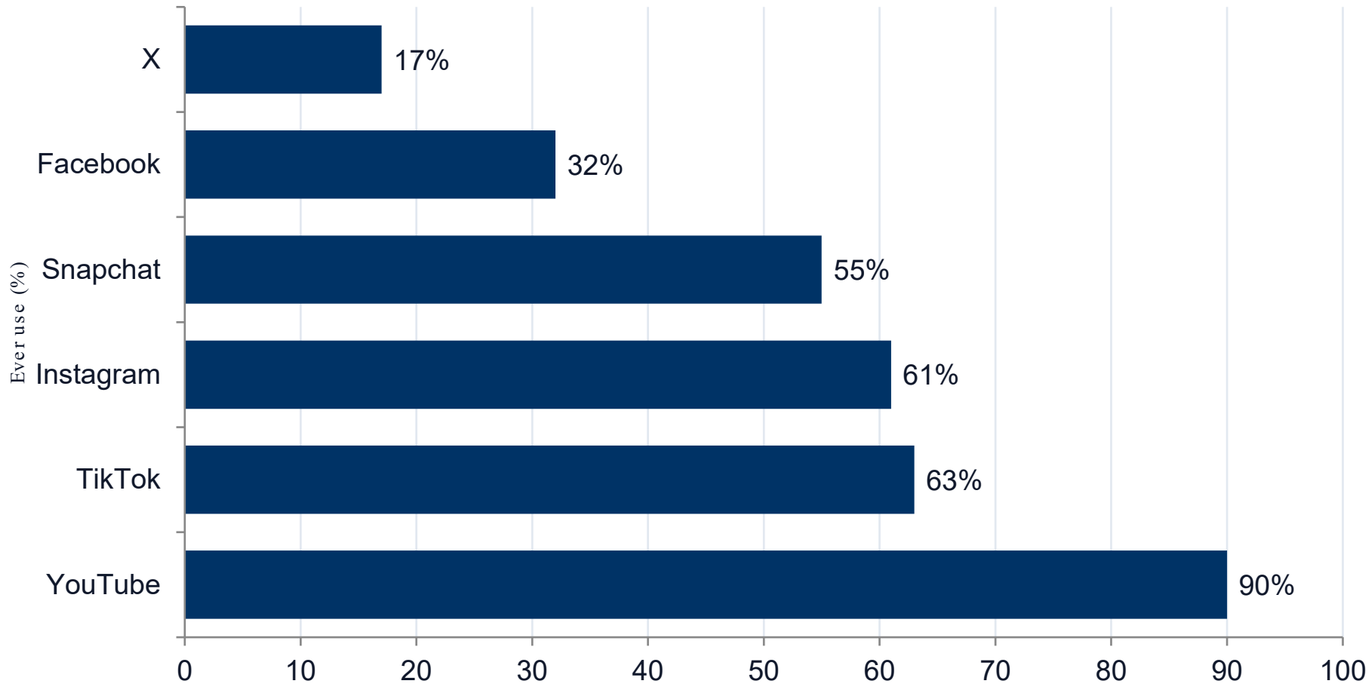
Sirui Han · HKUST
Lecture 8 · Week 9

Engineering choices shape what becomes visible, measured and governed.

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Social media is an AI - mediated environment

U.S. teens who say they ever use each platform



Interpretation boundary

High-use context does not establish an AI effect.
1,391 U.S. teens, ages 13–17
Fieldwork: 18 Sep –10 Oct 2024
Self-reported use; not a Hong Kong population estimate.

Source: Pew Research Center (2024) · pewresearch.org/internet/2024/12/12/teens-social-media-and-technology-2024/

AI operates across the platform stack

One platform can combine several AI functions, each with different inputs, decisions and accountabilities.

INPUTS · content · context · interaction logs · advertiser demand · policy rules

Creation
assist, translate, generate

Retrieval & ranking
select and order

Ads allocation
match, price, deliver

Moderation
detect, review, act

Analytics & governance
measure, audit, change

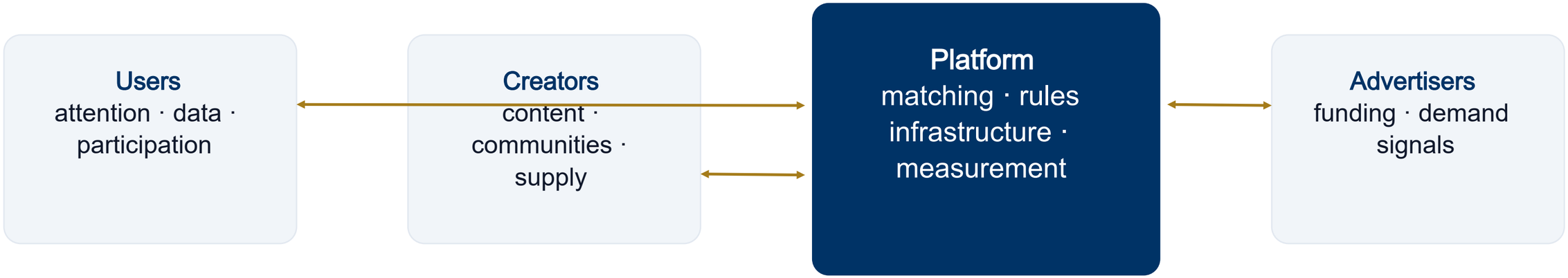
OUTPUTS · visibility · eligibility · distribution · enforcement · reported outcomes

Framework: functional system map for platform AI decisions

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The business model shapes the objective

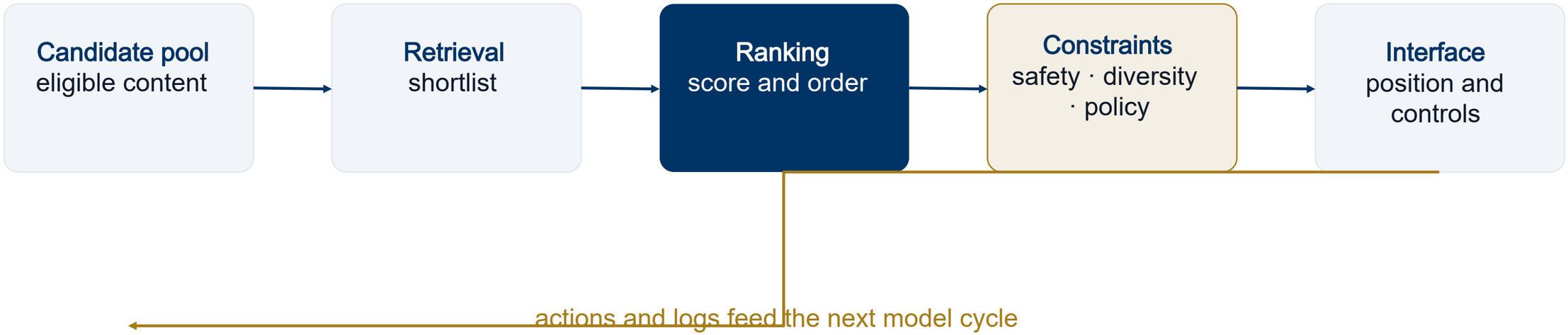
Platform AI mediates an exchange among participants; attention, revenue, trust and rights do not automatically align.



Revenue signal ≠ public - interest objective

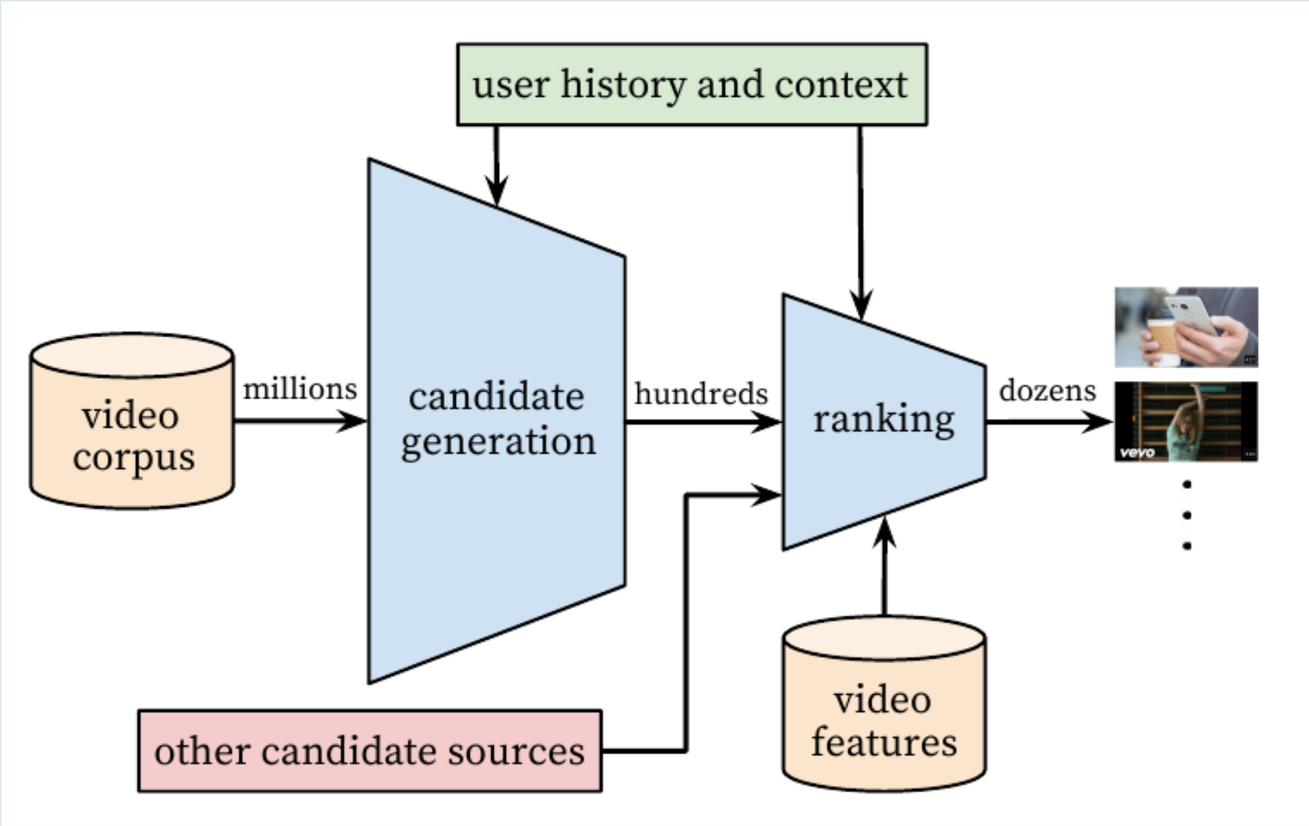
Recommendation is a staged decision pipeline

Selection, ranking and constraints each reshape the exposure a person receives.



Permitted signals, constraints and feedback channels determine the next decision.

YouTube: the two - stage architecture (2016)



Original Figure 2 · historical architecture

Candidate generation reduces a corpus from millions to hundreds.

Ranking scores the shortlisted items and serves dozens.

These are qualitative scale labels, not measured funnel conversion rates.

This 2016 paper does not describe a current 2026 service.

Source: Covington, Adams & Sargin, RecSys 2016, Fig. 2 · https://research.google.com/pubs/archive/45530.pdf

A better proxy can still be a proxy

Ranking objectives make some forms of user response legible and leave other consequences outside the score.

CTR
clicks / impressions
Captures entry, not sustained attention.

richer engagement signal →

Expected watch time
per impression
Captures more of what happens after a click.

Neither measure equals satisfaction, welfare or rights.

Source: YouTube ranking objective described in Covington et al. (2016) · https://research.google.com/pubs/archive/45530.pdf

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Responsible ranking is multi - objective

Proposed engineering frame: explicit trade - offs, evidence and control points replace a hidden single score.

Objective	Proxy or evidence	What can be missed	Proposed control
Utility	Predicted watch time / impression	Duration ≠ welfare	Satisfaction measures; holdouts
Safety	Classifier, review and appeal signals	Labels can be incomplete	Policy threshold; slice audit; rollback
Diversity	Candidate and exposure coverage	Variety ≠ fair access	Cohort monitoring
Privacy	Data use and retention	Consent can be unclear	Minimize signals; restrict access
Latency	Response time	Fast ≠ responsible	Service guardrail

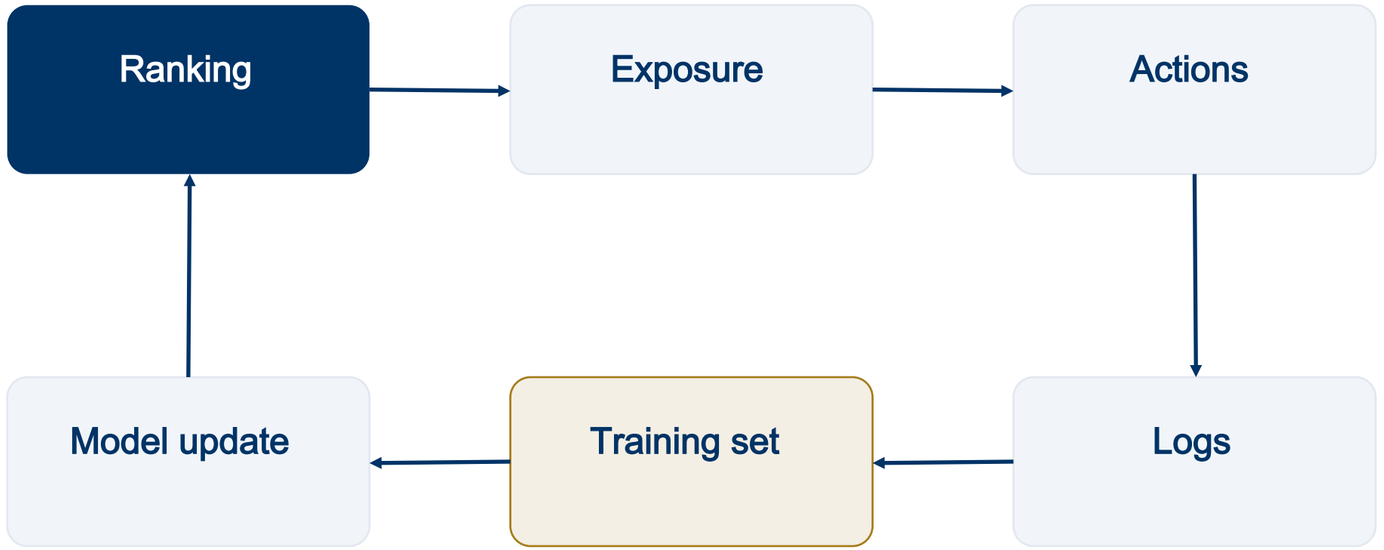
Conceptual proposal; proxy distinction motivated by Covington et al. (2016) · https://research.google.com/pubs/archive/45530.pdf

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Exposure changes the data used to personalize

Behavior logs reflect prior selection and interface placement before they become the next training signal.

Observed action
is shaped by position, network and context.



Creator adaptation
changes future candidates.

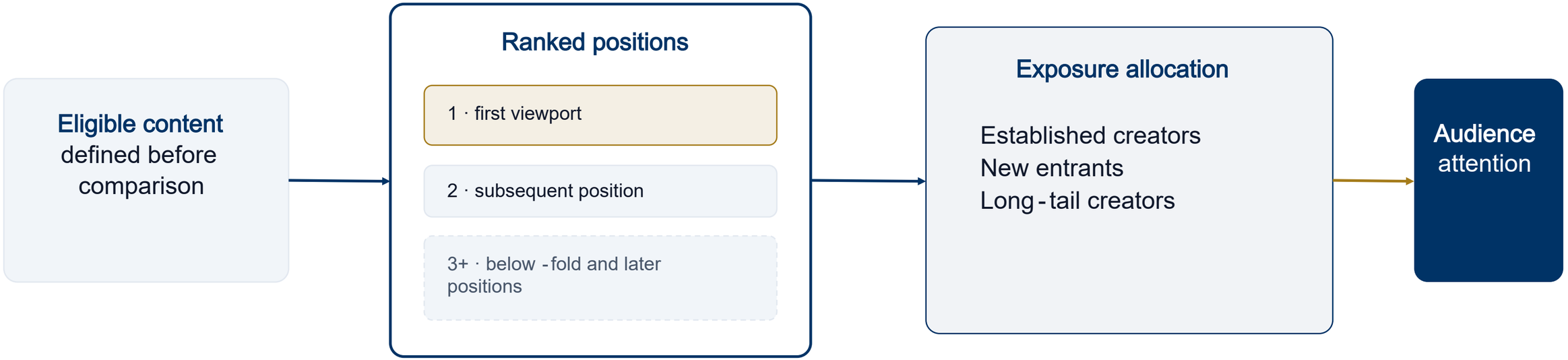
Proposed audits: position-aware logging · randomized exploration · delayed outcomes

Framework: conceptual feedback loop; logs are not neutral preference traces

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Visibility is an allocation decision

Ranking criteria allocate scarce high-attention positions; a metric can make concentration visible without deciding it away.



PROPOSED VISIBILITY CHECKS

impression share by cohort · top-k concentration · exposure conditional on eligible content

Framework: proposed creator-exposure monitoring; no platform distribution values asserted

Metric definitions determine the conclusion

A valid comparison states the numerator, denominator, population, time window and eligibility rule.

Metric	Numerator	Denominator	Interpretation boundary
CTR	Clicks	Shown impressions	Opening response, not completion
Watch time / imp.	Watched seconds	Shown impressions	Conditional on selection and position
Reach	Unique accounts exposed	Defined audience or population	Encountering content, not belief
Harmful exposure	Flagged exposed impressions	All served or sampled impressions	Exposure share, not incidence
Appeal error	Overturned appeals	Resolved appeals	Reviewed appeals, not all errors
Satisfaction	Positive survey responses	Completed or sampled responses	Respondent sentiment, not all users

Definition choices can change a conclusion without any underlying system change.

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Logged behavior is not pure preference

A behavioral log records action after a selected opportunity; it does not directly reveal a user's latent preference.



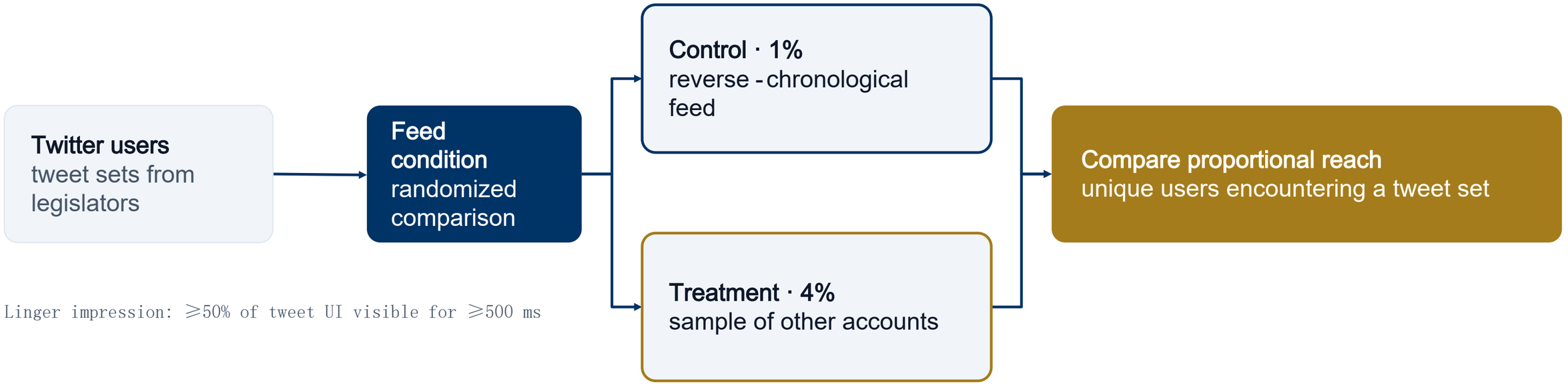
Confounding and selection enter before the log. Causal claims need explicit assumptions, randomized variation, or another cre dib le design.

Course causal model · exposure, position, network, and context are distinct inputs to observed behavior

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Twitter: a randomized exposure comparison

Analysis of 3,634 legislator accounts across seven selected countries, 1 April –15 August 2020.



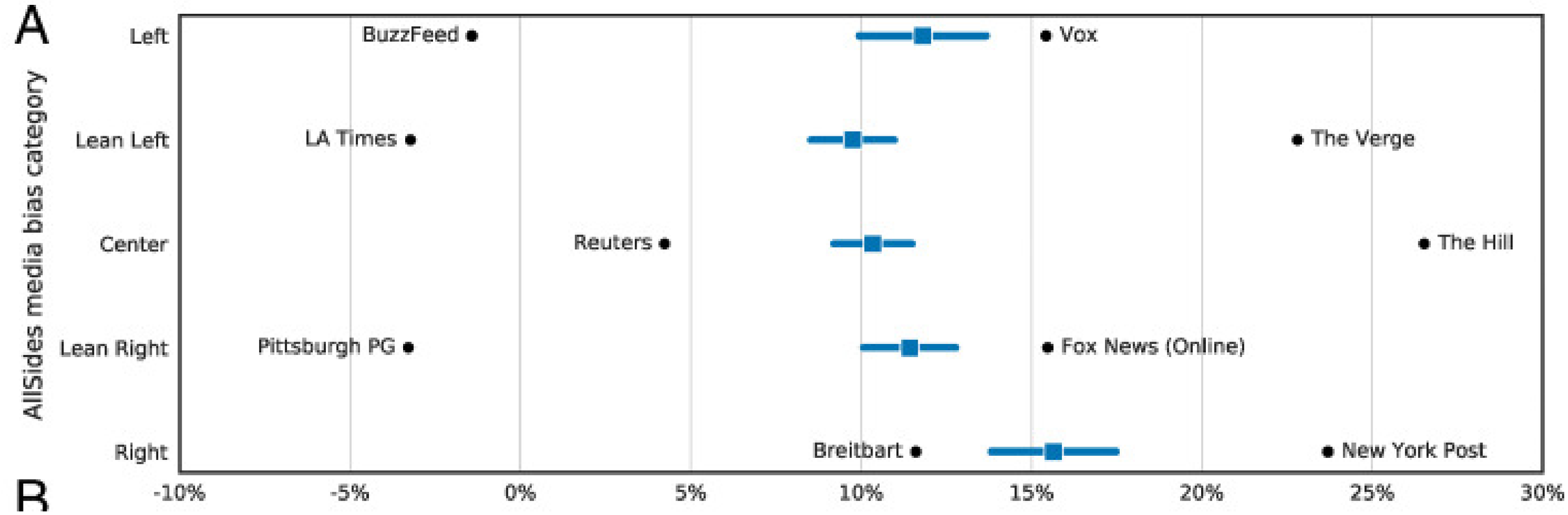
Linger impression: ⩾50% of tweet UI visible for ⩾500 ms

Amplification = $[(r_t + 1) / (4r_c + 4) - 1] \times 100\%$ — sampling fractions are adjusted in the comparison

Source: Huszár et al., PNAS (2022) · historical platform experiment

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Amplification is a relative - reach measure



Blue = group mean ± bootstrap SD

Black dots = individual sources

Relative exposure (proportional reach), not votes, attitudes, or causal persuasion.

Original Figure 2A · Huszár et al., PNAS (2022)

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Exposure, attitudes, and behavior are different outcomes

Inference weakens when a result at one rung is used to claim an unmeasured downstream outcome.

1 · Exposure
What entered the visible feed
Reach · impressions · position

→

2 · Interaction
Attention or action on what was shown
Linger · click · share

→

3 · Attitudes
Beliefs, knowledge, participation
must be directly measured

→

4 · Behavior / welfare
Context-dependent downstream outcomes
not inferred from clicks

Experimental bounds matter: interference, changing treatment, opt-outs, and selected countries can limit both the estimated effect and generalization.

Source context: Huszár et al., PNAS (2022) · outcome ladder is course analysis

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Meta: replacing the algorithmic feed

Preregistered randomized U.S. election field experiment, 24 September –23 December 2020; published in 2023.

Facebook · n = 23,391
U.S. adults · status-quo algorithmic ↔ chronological

Instagram · n = 21,373
U.S. adults · status-quo algorithmic ↔ chronological

→

Measured during ~3 months
Exposure mix
Time and engagement
Political attitudes, knowledge, participation

→

Observed result
No detectable statistically significant effect on measured political attitudes, knowledge, or participation during the study.

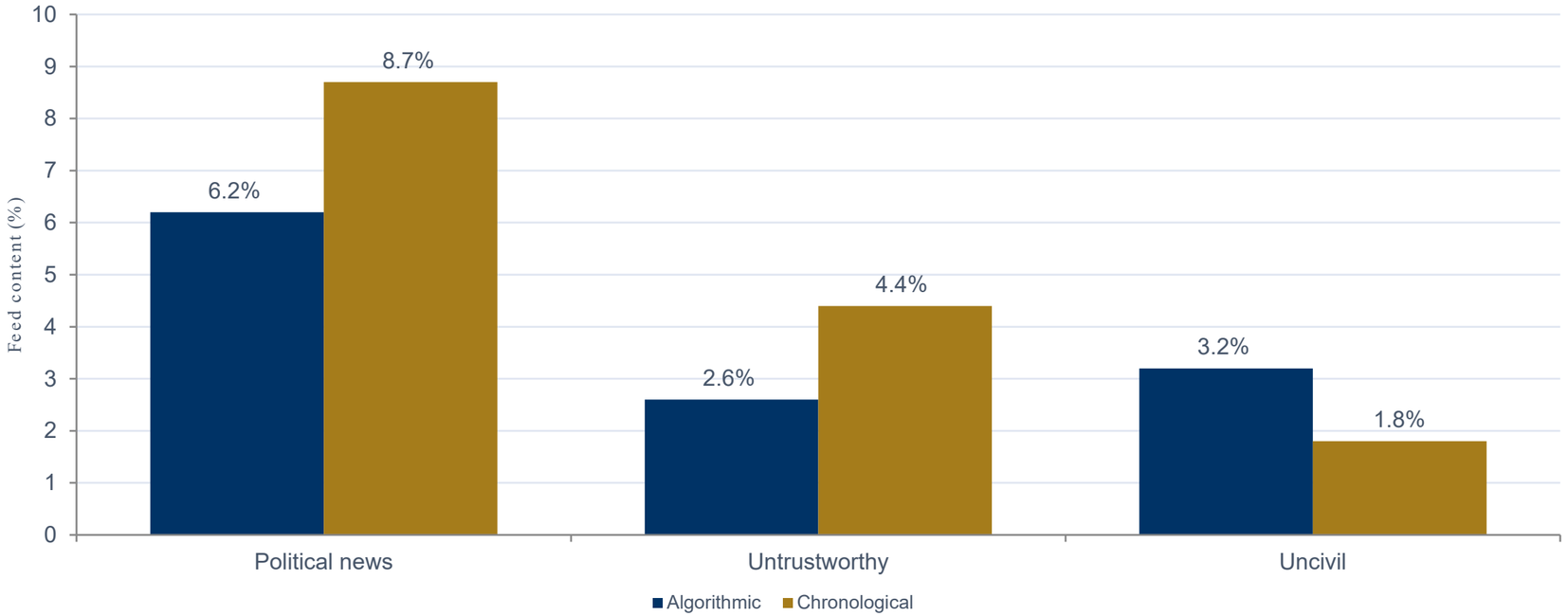
Bounds: consenting, relatively active users; one election period; specific feed treatment; ads unchanged.

Source: Guess et al., Science (2023) · experiment fielded in 2020

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Chronology changes exposure in mixed directions

Facebook only · average unweighted shares of measured feed content



Read the denominator
Percent of measured feed content
Not percent of users
Not an estimate of platform-wide prevalence

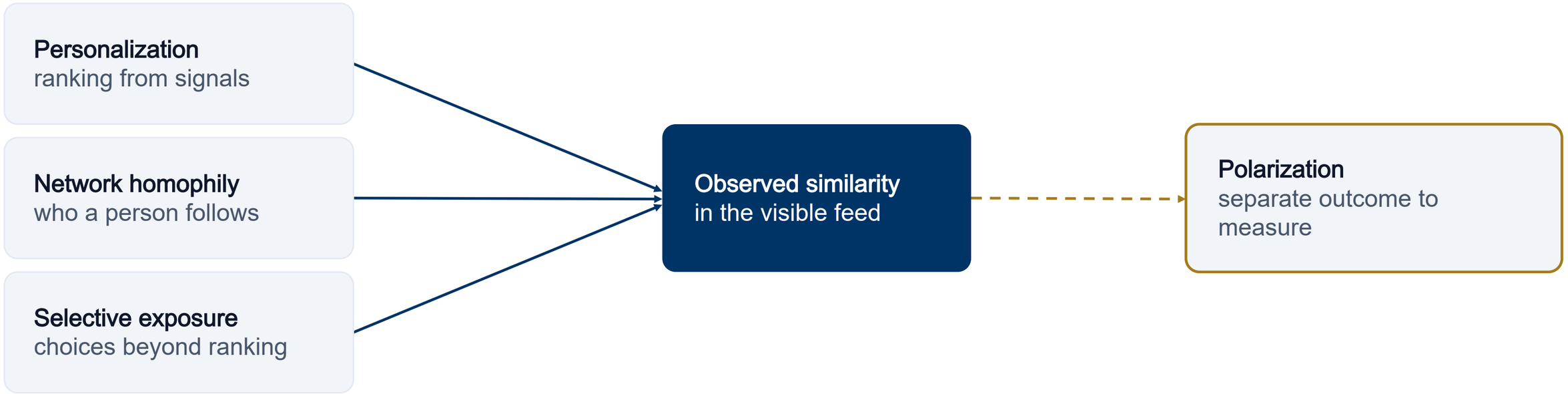
A chronological feed is not uniformly lower or higher on these measures.

Randomized treatment and time window are study-specific; chart excludes Instagram data.

Source: Guess et al., Science (2023) · Facebook feed-content shares

A filter bubble is a hypothesis, not a universal law

Observed similarity in a feed can arise through multiple mechanisms; personalization is only one of them.



The ~3 - month Meta field experiment found no detectable effect on measured political attitudes, knowledge, or participation; that is not a universal null.

Source context: Guess et al., Science (2023) · mechanisms diagram is course analysis

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Personalization and invisible editorial choices



Eli Pariser
Beware online filter bubbles (TED, 2011)

Full source video

Historical argument and anecdotes, not causal evidence.

Engineering focus: make selection rules visible, contestable, and auditable.

Source: TED Talk (2011) · conceptual figure is course analysis

Feed controls are interventions with trade - offs

Control	Design intent	Measurement effect / trade - off
Chronological option	Restore a time - ordered view	May change freshness, coverage, and exposure mix; not automatically safer
Preference reset	Give users a way to interrupt inferred history	Needs clear state, reversible defaults, and a check for unequal utility loss
Diversity constraint	Protect candidate breadth or viewpoint coverage	Requires a definition, slice metrics, and monitoring for relevance loss
Exploration	Test beyond predicted top items	Creates exposure risk; bound it with eligibility and safety constraints
Explanation	Make a selection reason legible	Can improve contestability, yet may oversimplify or expose sensitive signals

Proposed engineering controls, not claims about any platform’s current design.

Course design analysis · every control needs an owner, evidence threshold, and rollback path

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Benefits and harms depend on the system and context

Ethical lens	Potential benefit	Potential harm	Engineering evidence / control
Autonomy & agency	Relevant discovery and usable controls	Opaque steering or burdensome defaults	Choice architecture, explanation quality, reset and appeal use
Expression & participation	Voice, connection, and distribution	Suppression, harassment, or unequal access to attention	Exposure slices, moderation error analysis, contestability
Fairness & visibility	Broader candidate coverage	Feedback loops and arbitrary creator disadvantage	Ranking audits by cohort, coverage, and position
Welfare, safety & privacy	Supportive communities and safer discovery	Harmful exposure, data leakage, or excessive collection	Safety constraints, minimization, incident response, rollback

A benefit in one context does not erase a harm in another; evaluate the deployed system, population, and evidence.

Course ethical matrix · not a claim that any platform or feed has a uniform effect

PART II · CONTENT GOVERNANCE AND GENERATED MEDIA

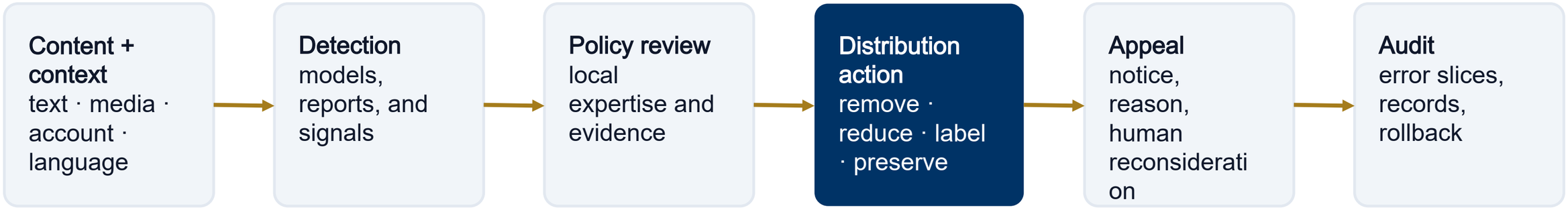
Intermission

From measurement to operational responsibility

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Moderation combines policy, models, and operations

Detection proposes an action; policy, local review, appeal, and audit make the system governable.



A classifier alone cannot establish legitimacy: decisions require policy boundaries, accountable operations, and evidence that correction works.

Moderation errors have asymmetric costs

The relevant outcome is policy consistency, not a universal classifier score.

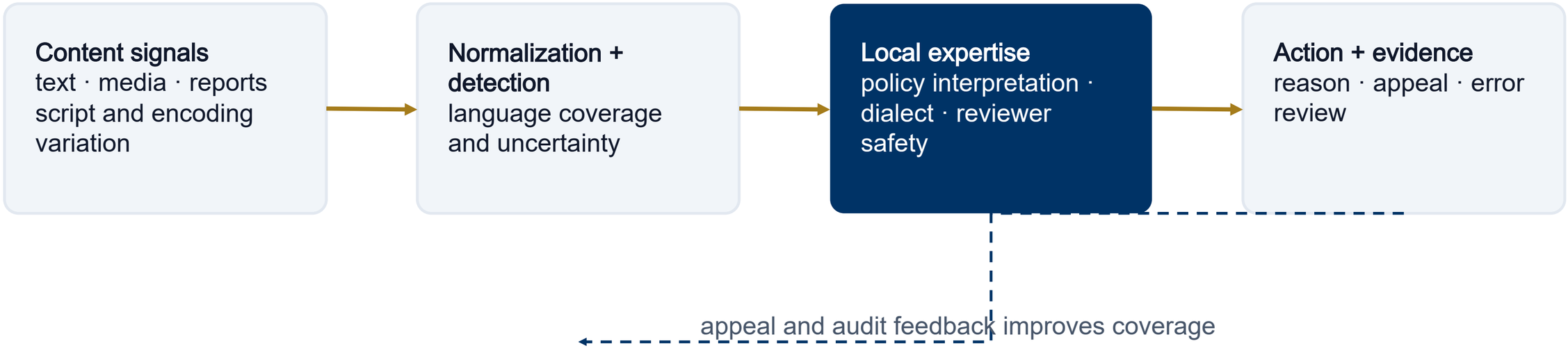
Decision	Policy-violating	Permitted
Act	**True positive** Enforce the policy with a reasoned record.	**False positive** Permitted expression is suppressed; appeal burden rises.
Allow	**False negative** Potentially violating content remains exposed.	**True negative** Permitted expression remains available.

Thresholds encode trade-offs among speech, safety, local context, review capacity, and meaningful appeal access.

Conceptual cost matrix · error rates alone do not determine the decision threshold

Language and context are part of model quality

Coverage depends on scripts, encodings, dialect, coded language, and timely local knowledge.



Local expertise is a system capability: it changes both the interpretation of evidence and the quality of decisions.

Myanmar: platform scale meets local - context gaps



BSR Human Rights Impact Assessment · October 2018

- Documented script and language - context constraints, including Zawgyi and Unicode normalization.
- About 20 million Facebook users in Myanmar in mid - 2018 is a platform - scale estimate, not a harm count.
- Myanmar's conflict involved political, military, and social forces; the platform is not a sole cause.

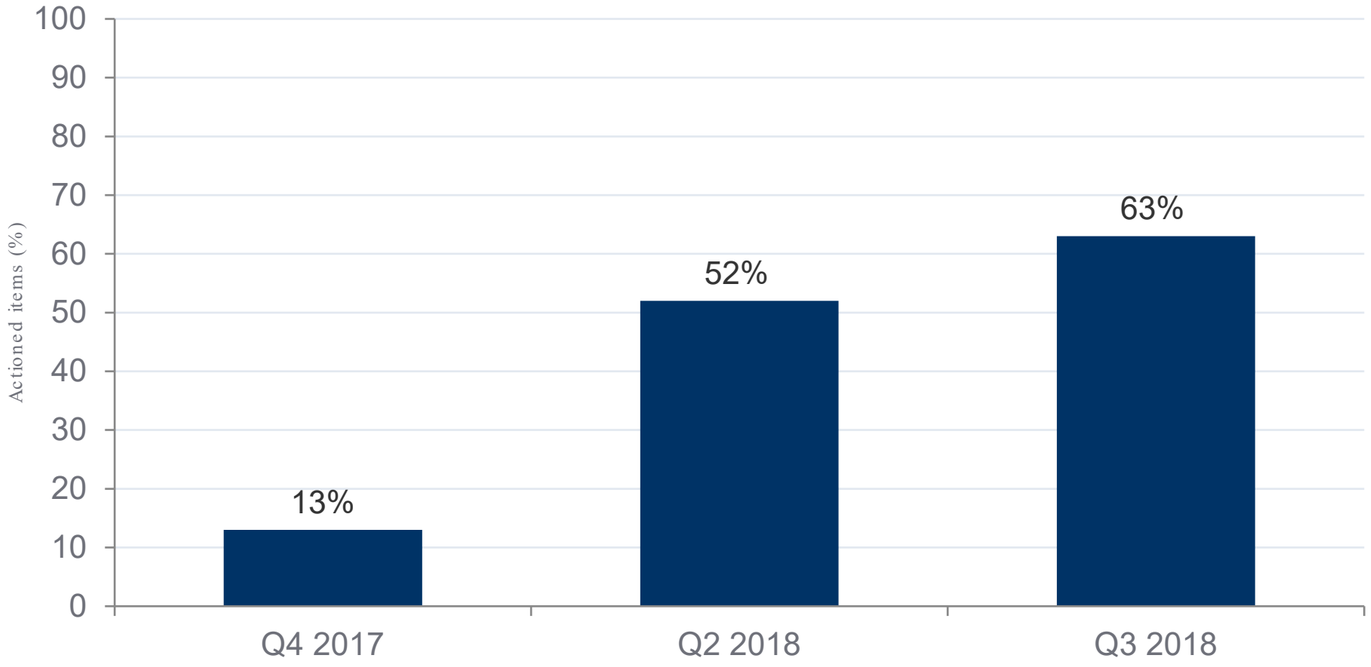
Commissioned by Facebook; BSR stated editorial control. A company response is not an independent audit.

Source: BSR, Human Rights Impact Assessment: Facebook in Myanmar (2018)

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An enforcement percentage is not a safety outcome

Company-reported proactive identification among actioned Myanmar hate-speech items



Denominator: actioned items
not all posts or all harmful content

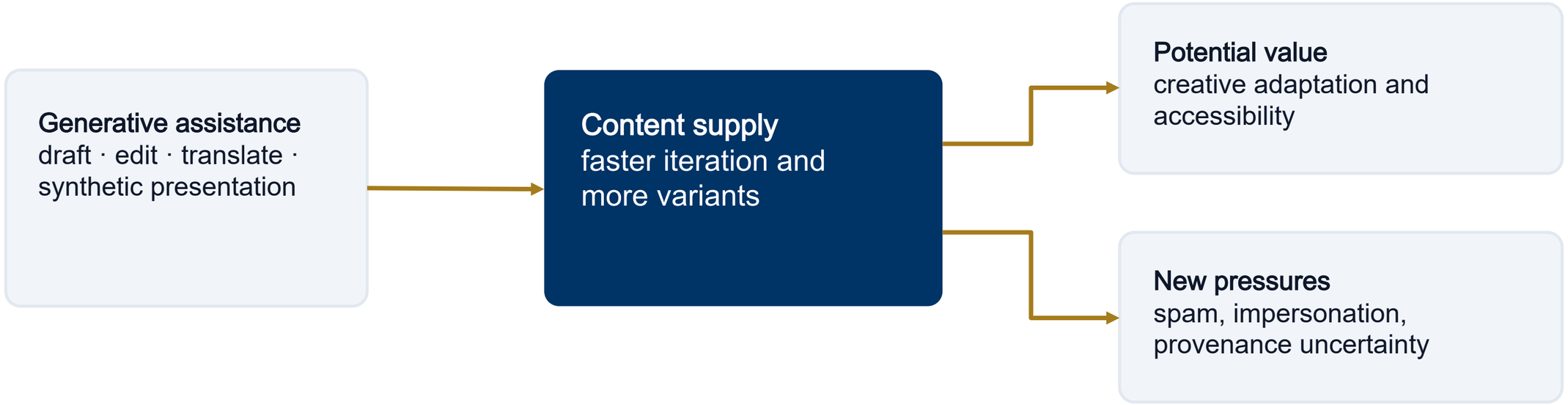
Q3 2018: about 64,000 actioned items.
Comparable totals were not reported for the other quarters.

Not recall, precision, prevalence, or an offline safety measure.

Source: Facebook response to Myanmar HRIA (5 Nov 2018) · company-reported metric

Generative tools alter the content supply

Automation changes volume, variety, and provenance signals before ranking systems encounter an item.



Responsibility spans tool design, creator choices, platform distribution, and the evidence attached to an asset.

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The fabricated Pentagon image (2023)

FALSE / FABRICATED



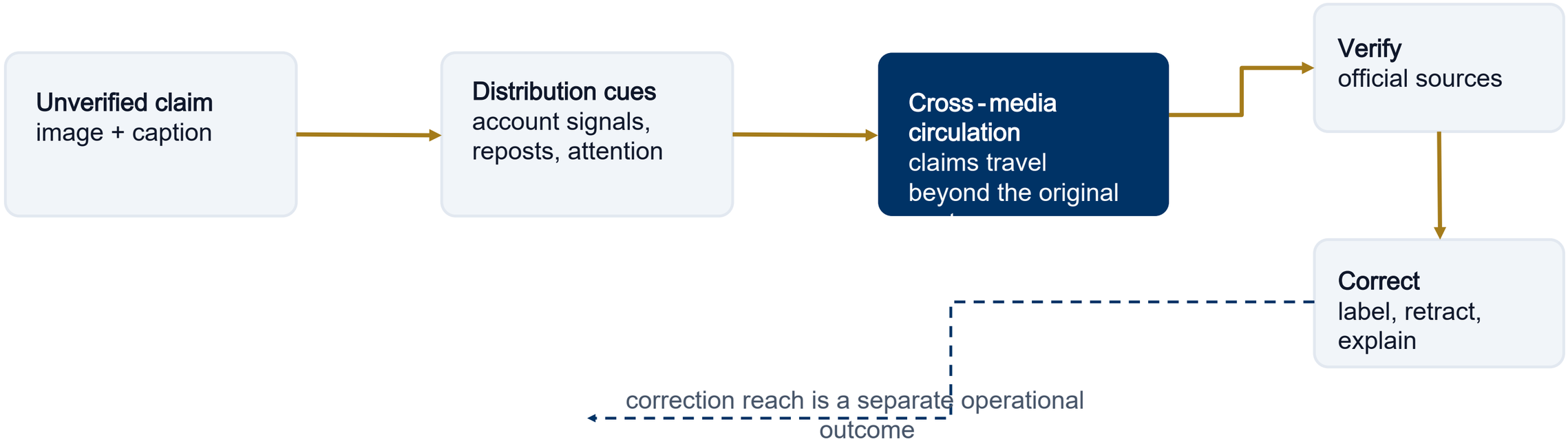
Official denial: no explosion or incident near the Pentagon.
Reuters verdict: the circulating claim was false.
Provenance limit: AI origins were suspected, not independently proved.

Event facts ≠ source identity ≠ image provenance.

Image source: SMH analysis (2023) · Fact check: Reuters

Credibility cues can accelerate a false claim

Payment verification badges identify an account status; they do not authenticate a claim.



Conceptual incident pathway · Sources: CNN and Reuters

Provenance and correction need layered controls

No single label resolves truth, coverage, bypass, or the reach of a correction.

1 · Creation — Capture origin information, preserve context, and make provenance signals inspectable.

↓

2 · Distribution — Attach context, apply proportionate friction, and preserve reporting and escalation paths.

↓

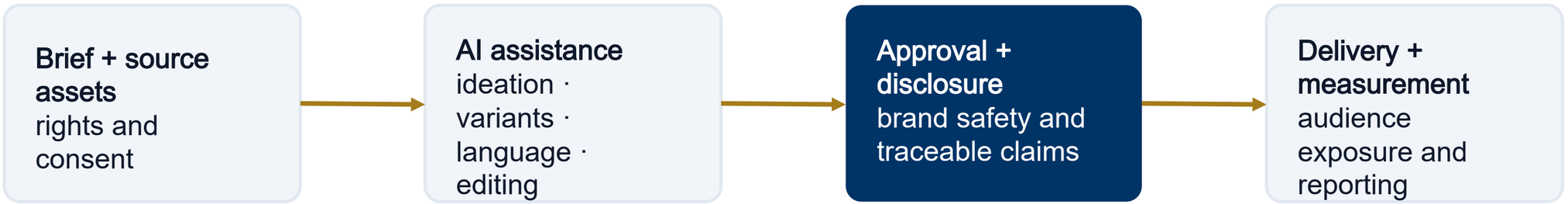
3 · Incident response — Verify with authoritative evidence, record decisions, and measure whether corrections propagate.

Proposed engineering control stack · provenance is evidence about origin, not proof of truth

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AI changes creator and advertising workflows

Assistance upstream of delivery changes who must establish rights, disclosure, and accountability.



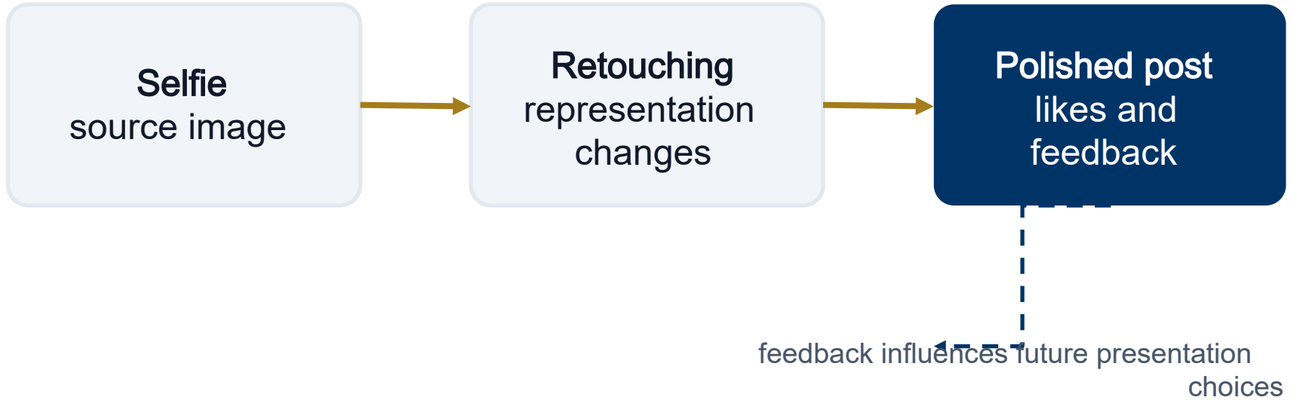
Decision rights travel with the workflow: creator, advertiser, platform, and tool provider have distinct responsibilities.

This is an illustrative control map, not a claim about any platform's targeting algorithm.

Reverse Selfie: constructed identity and feedback



Retouching and feedback are connected design choices



The film reverses edits applied to a teenage girl's selfie. It illustrates retouching and feedback, not proof of a specific AI tool or clinical harm.
Source film: Full source video

Campaign source: WPP / Ogilvy, Dove Reverse Selfie case study (2022)

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Youth well - being requires more than engagement metrics

Well-being is mediated by people, content, social context, design, and time —not a single engagement score.

Potentially supportive pathways

Connection and community
Creative expression and identity exploration
Access to information and support

Potentially harmful pathways

Sleep displacement and social comparison
Exposure to harmful content or harassment
Problematic patterns of use

Mechanisms and associations inform safeguards; they do not establish AI -specific clinical causation.

Source: U.S. Surgeon General advisory, 2023

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TikTok Lite Rewards: a product - governance case

European Commission Digital Services Act case · France and Spain · 2024



Formal proceedings opened
Risk assessment for the rewards programme

Voluntary suspension
Rewards programme paused in the EU

Binding withdrawal
Permanent EU withdrawal and anti-circumvention commitment

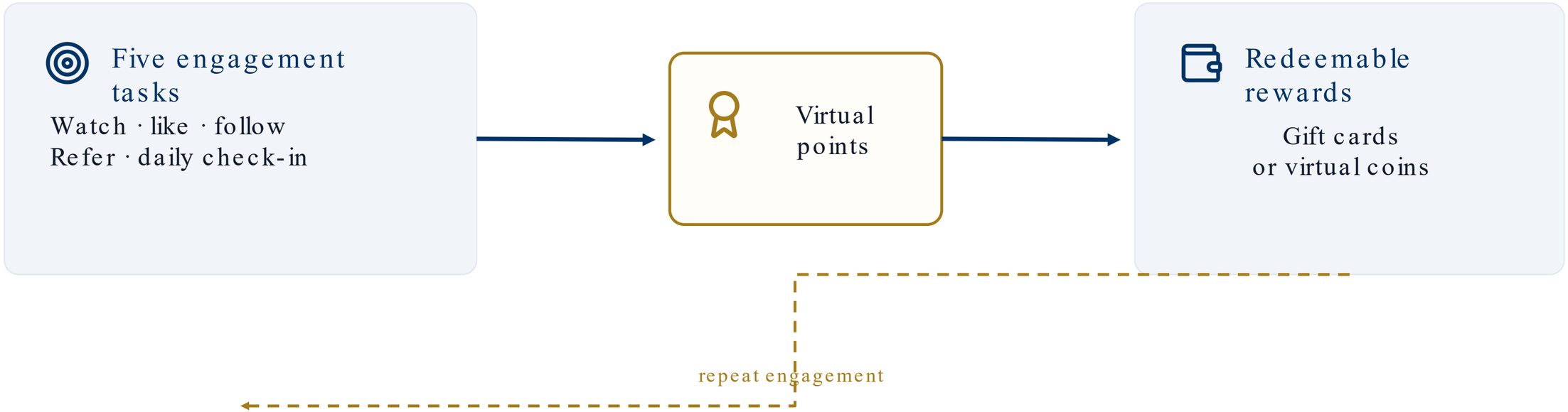
Resolved through commitments; no admitted infringement. This is a feature - risk case, not a diagnosis of harm.

Sources: Proceedings · Binding commitments

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The reward loop is a design choice

ENGAGEMENT INCENTIVE LAYER ≠ CORE RECOMMENDER FINDING



Official materials verify the mechanics—not a user denominator, effect size, or medical outcome.

Source: European Commission TikTok Lite commitments, 2024

A responsible dashboard separates outcomes

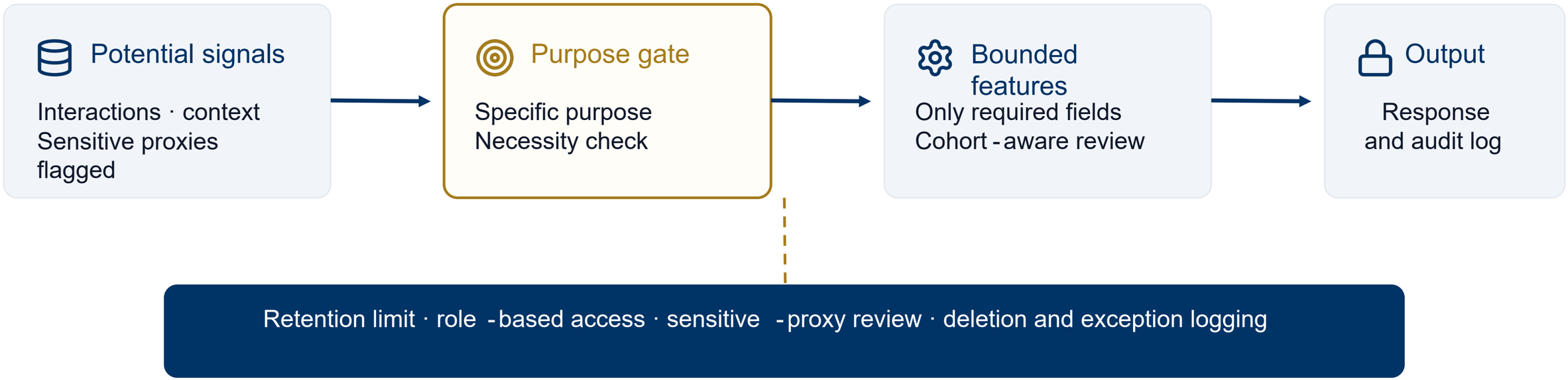
Proposed measurement design · values are deliberately absent

Outcome family	Signal and denominator	Accountable owner	Decision use
Engagement quality	Session / return pattern (not welfare)	Product analytics	Investigate proxy drift
User agency	Settings uptake · opt-out completion	Product and UX	Escalate friction
Harm exposure	Policy-violating exposure rate	Safety science	Tighten distribution
Moderation quality	Appeal overturn rate	Trust and Safety operations	Review error slice
Data footprint	Retention exceptions · sensitive-proxy access	Privacy engineering	Block unapproved use
Business health	Creator retention · support burden	General manager	Do not trade off gates

Proposal: define outcomes, denominators, owners, and escalation paths before collection.

Data minimization limits the personalization footprint

Privacy is a design constraint at every stage, not only an access-control setting.



Proposal: minimization limits collection, inference, retention, and downstream reuse.

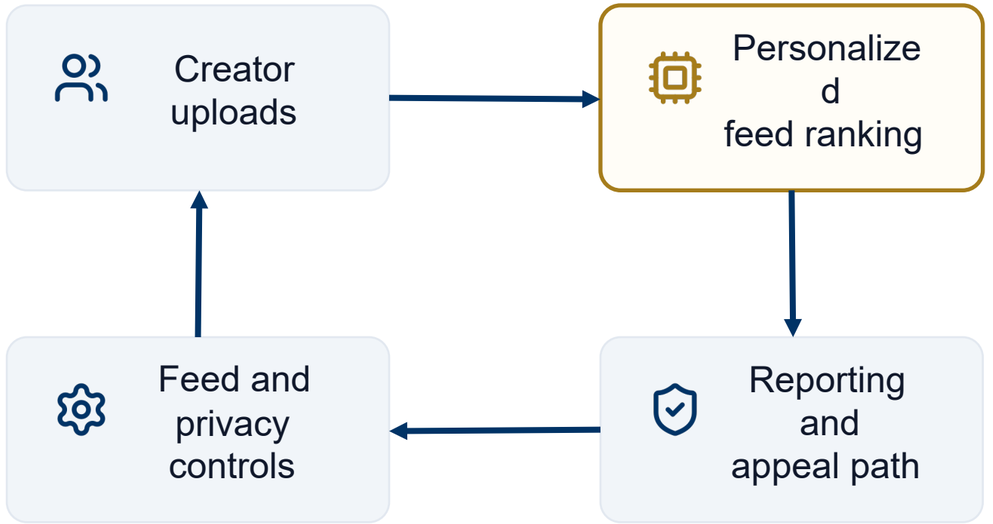
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A short - video launch: the decision context

FICTIONAL COURSE SCENARIO · NO REAL PLATFORM OR PERFORMANCE DATA

LoopLine

Hypothetical short - video service
Proposed setting: older - teen cohort
Launch decision concerns feature scope and safeguards



Feature assumptions

Opt - in personalization
Chronological alternative
Creator feedback signals
Moderation escalation

Fictional classroom case: feature assumptions only; no numerical forecasts, metrics, or synthetic dataset.

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Launch conditions and decision rights

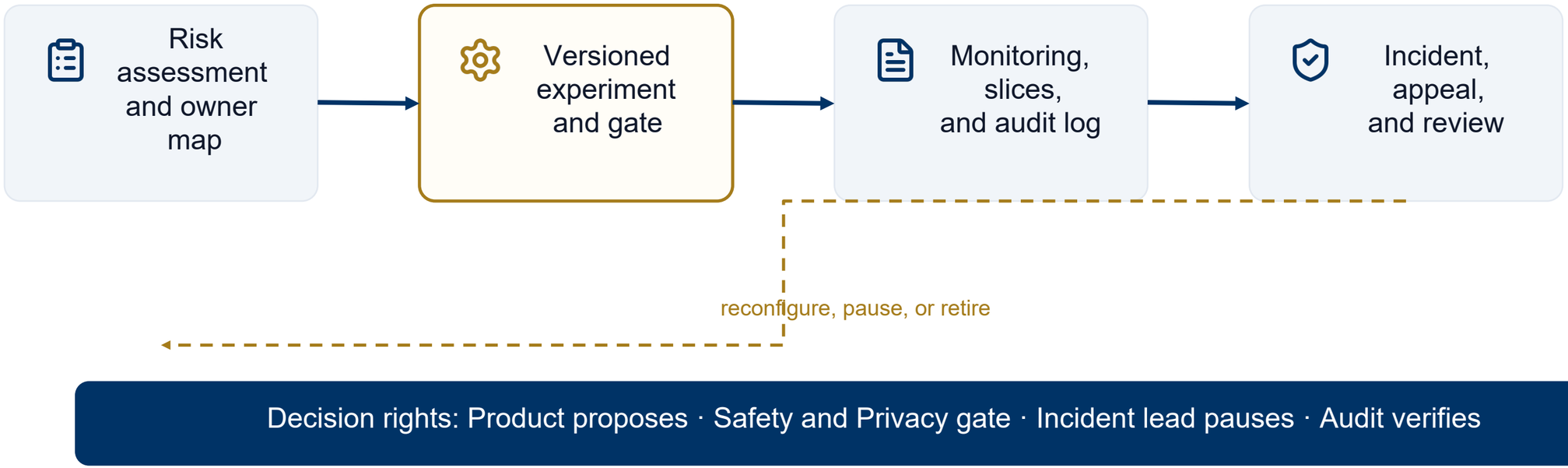
Fictional LoopLine case · proposed launch gates, not a record of platform practice

Launch condition	Evidence artifact	Control	Accountable owner	Rollback condition
Agency / choice	Pre-launch usability record for settings and chronological option	Prominent controls; default audit	Product lead	Material control failure
Privacy purpose	Data inventory and retention register approved	Minimize inputs; access logs	Privacy engineering	Unapproved purpose or access
Harm pathways	Scenario review and escalation protocol	Distribution caps; human queue	Safety lead	Severe incident pattern
Reversibility	Versioned experiment and holdout decision record	Feature flag; freeze or rollback	Incident owner	Gate threshold crossed

Fictional applied matrix: condition, evidence, control, owner, and rollback are coupled design decisions.

Governance closes the engineering loop

Governance outputs evidence trails and decision rights —not only an approval checkpoint.



Proposed operating model for accountable experimentation and incident response.

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Design for evidence, agency, and reversibility



Proxy ≠ welfare • reach ≠ belief • enforcement rate ≠ safety • provenance ≠ truth

Synthesis of dated cases; these distinctions resist unsupported universal causal claims.

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Case evidence and original figures

Primary records and original figures used in the case analysis

Case	Short reference	Use in lecture	Primary link
YouTube	Covington et al., RecSys 2016	Figure 2 / historical two-stage architecture	Paper · ACM record
Twitter	Huszar et al., PNAS 2022	Original Figure 2 / relative reach	PNAS paper
Meta feeds	Guess et al., Science 2023	Facebook feed experiment	Science paper
Myanmar	BSR Human Rights Impact Assessment, 2018	Language and operations context	BSR report
TikTok Lite	European Commission DSA, 2024	Proceedings and commitments	Commitments
Pentagon image	Reuters Fact Check, 22 May 2023	False claim and official denial	Reuters

URLs are embedded in concise, labelled links; citations are dated to preserve scope.

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Media and further reading

Media provenance

Eli Pariser, TED 2011
Historical argument about personalization, not causal evidence.
Full source video · Official YouTube

Dove Reverse Selfie, 2021 campaign
Representation and feedback artefact, not proof of clinical harm.
Full source video · Agency case

Evidence boundaries and reading

Pew 2024 — 1,391 U.S. teens aged 13–17; dated survey, not Hong Kong or AI causation.
Report

HHS 2023 advisory — benefits and risks; mechanisms and associations, not AI-specific causation.
Advisory

Course text: O'Neil, *Weapons of Math Destruction*

Transferability is bounded by population, outcome, source type, and study design.